

Campus HelpDesk

Project Proposal and Design Document

Project information	Details
Project domain	Campus ticketing and support
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1. Project Domain

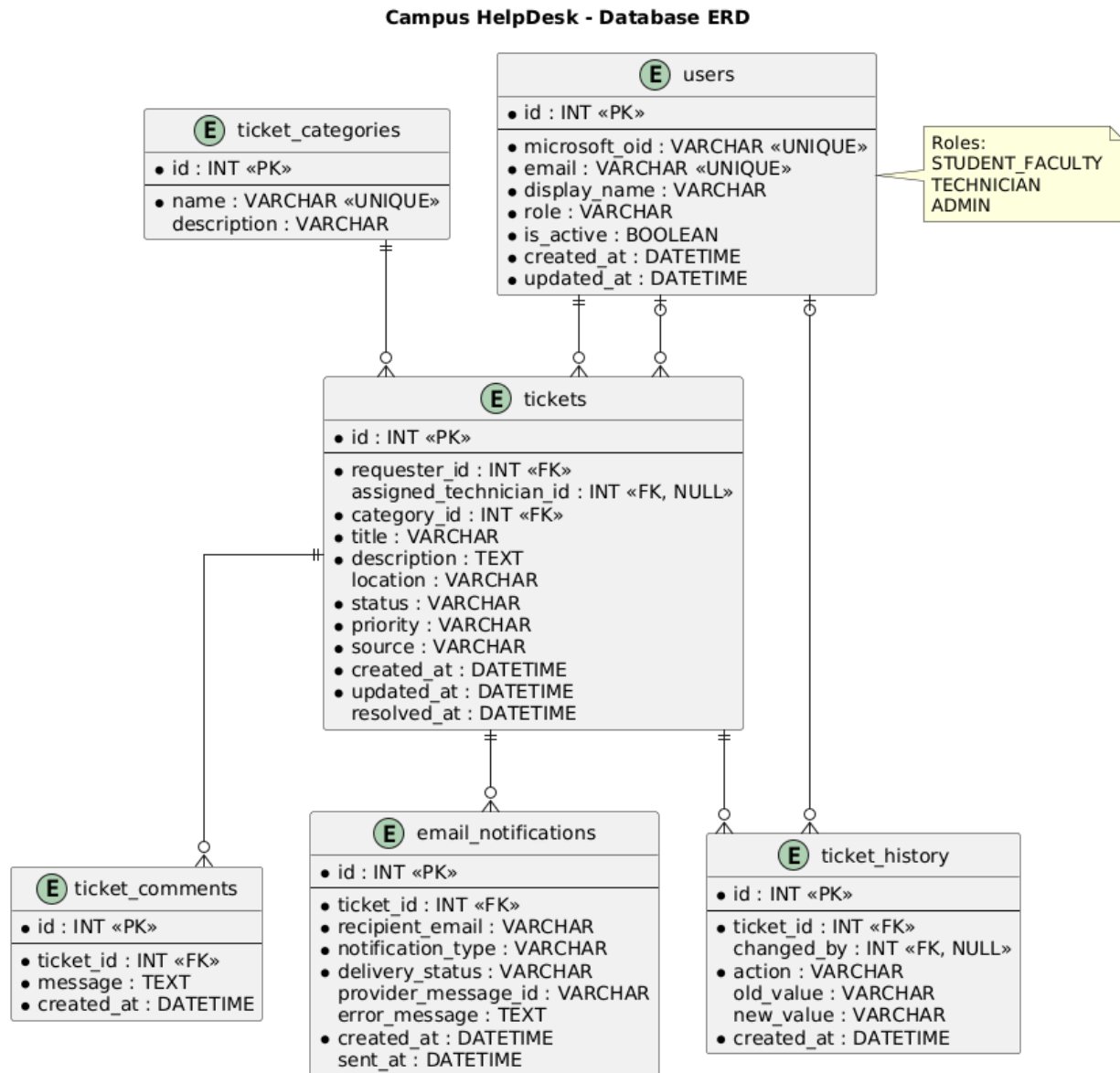
Campus HelpDesk is a web system for university students and faculty to report problems. Technicians can claim, update, and resolve tickets, while administrators can manage the whole system.

Main Functions

Function	Description
University login	Users sign in with university credentials
Ticket management	Users create tickets; technicians assign, update, comment on, and resolve them.
Email notification	Users receive email when a ticket is created, assigned, updated, or resolved.
Peer integration	The HelpDesk exchanges ticket and enrollment data with another team's backend.

2. Database Schema (ERD)

The database uses MySQL or PostgreSQL through Prisma ORM and Prisma migrations.



Database Tables

Table	Purpose
users	Stores AD identity, email, name, and system role.
ticket_categories	Stores selectable ticket categories.
tickets	Stores the reported problem, status, priority, requester, and technician.
ticket_comments	Stores comments belonging to a ticket.
ticket_history	Stores changes made to a ticket.
email_notifications	Stores email delivery attempts and their status.

3. RBAC Roles

Role-Based Access Control limits what each user can do.

Role	Permissions
Student / Faculty	Create tickets, view their own tickets, and add comments.
Technician	View available tickets, claim tickets, update status, and resolve tickets.
Administrator	View and manage all tickets, users, categories, and technician assignments.

4. API Integrations

Third-Party API: Brevo Email API

The backend will use Brevo to send ticket notifications. Email is an extra service; ticket operations will still succeed if an email cannot be sent. The Brevo API key will be stored in Azure Key Vault.

Peer API: Course Registration (EduCore)

Direction	Planned endpoint and use
Expose	POST /peer/tickets — EduCore can create a HelpDesk ticket when registration fails.
Consume	GET /peer/enrollments/:studentId — HelpDesk retrieves course details for registration-related tickets.

Direction	Planned endpoint and use
Security	Both backends authenticate with an x-api-key header. Keys are stored in Azure Key Vault.